**DH Software Exercise Assignment**

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**Part 1: Continue with Lancsbox X**

***1a) Using the KWIC tool, give both the relative and raw frequencies of your selected word used each by Dickens and Austen.***

Using the KWIC tool in Lancsbox, I searched for the word “family” in Dickens and Austen’s corpora.

The raw frequency of family in the Dickens corpus was 1456 tokens, and the raw frequency of family in the Austen corpus was 611 tokens. The relative frequency per one million tokens was 337.00 for Dickens and 774.89 for Austen. This indicates that Austen uses the word family more frequently than Dickens, but Dickens uses the word family more often than Austen because the raw frequency of family in the Dickens corpus was higher than in Austen.

***1b) Use the KWIC tool to query adjective + your_word, following the example of ADJECTIVE[word="your_word"]. Copy and paste 20 lines from the output for each author into your assignment document. Describe any patterns you find.***

Dickens:

| | | | |
|---|---|---|---|
| Little_Dorrit.txt | took it ill if any | other family | had much to say to |
| Little_Dorrit.txt | The Barnacles were a very | high family, | and a very large family. |
| Little_Dorrit.txt | high family, and a very | large family. | They were dispersed all over |
| Little_Dorrit.txt | children. The ages of the | whole family | he made known to Mr |

| | | | |
|---|---|---|---|
| Little_Dorrit.txt | lock, and to bear a | horrible family | resemblance to Mrs Bangham--he |
| Little_Dorrit.txt | as gratified as if his | whole family | were there. All the natural |
| Little_Dorrit.txt | was the head of the | fallen family; | and bore, in her own |
| Little_Dorrit.txt | table the flavour of the | whole family- | tree. In its presence, his |
| Little_Dorrit.txt | the heraldic cognisance of the | old family | to whom the property had |
| Little_Dorrit.txt | as to say, in his | private family | circle, that he feared Mr |
| Little_Dorrit.txt | friendly terms. There was a | little family | misunderstanding among us. Your respected |
| Little_Dorrit.txt | for the demand. That high | old family, | the Barnacles, had long been |
| Little_Dorrit.txt | with his young wife and | little family, | at his Cure of Souls. |
| Little_Dorrit.txt | proving to be of a | wealthy family, | and having come into the |
| Little_Dorrit.txt | that very high and very | large family | might shed as much lustre |
| Little_Dorrit.txt | Merdle’s, harped at their | own family | breakfast-table on his wish |
| Little_Dorrit.txt | while the whole of his | young family | shook the table, and what |
| Little_Dorrit.txt | after six lessons when the | young family | was under restraint. The tenancy |
| Little_Dorrit.txt | sighed Little Dorrit, with the | whole family | history in the sigh. ‘Yes. |
| Little_Dorrit.txt | To the health of | distinguished | of the fair ladies, your |

| | | | |
|---|---|---|---|
| | your | family-- | |

Through observation of the 20 lines from the output of the Dickens corpus, the word “family” contains social and class meaning. He often uses adjectives that describe someone’s social status, such as “old family”, “wealthy family”, “fallen family”, “high family”, and “distinguished family”. Additionally, certain adjectives reflect family size, such as “whole family”, “large family”, and “little family”. “Little family misunderstanding”, “private family circle” show family relationships. Connecting to the context and the social background of Dickens, “family” not only represents individuals’ origin, but it also reflects social hierarchy and criticizes society.

Austen:

| | | | |
|---|---|---|---|
| Emma.txt | wants and sufferings of the | poor family, | however, were the first subject |
| Emma.txt | Emma. “I had only my | own family | to study from. There is |
| Emma.txt | I need not bring the | whole family, | you know.” Here Mr. Weston |
| Emma.txt | Highbury, and born of a | respectable family, | which for the last two |
| Emma.txt | younger branch of a very | ancient family— | and that the Eltons were |
| Emma.txt | she would get into any | great family. | How did you think the |
| Emma.txt | large, as it included one | other family, | a proper unobjectionable country family, |
| Emma.txt | with great animation of a | large family | of young ladies that his |
| Emma.txt | over. The affection | whole family, | the warm attachment |

| | | | |
|---|---|---|---|
| | of the | | of Miss |
| Emma.txt | to pay to a poor | sick family, | who lived a little way |
| Emma.txt | even excluded from Hartfield. The | delightful family | party which Emma was securing |
| Mansfield_park.txt | hope of being with his | beloved family | again early in November. Maria |
| Mansfield_park.txt | notice of many of her | own family | likewise. She had loved, she |
| Mansfield_park.txt | his decided preference of a | quiet family | party to the bustle and |
| Mansfield_park.txt | indifference. They were a remarkably | fine family, | the sons very well-looking, |
| Emma.txt | cameos, corals, shells, and every | other family | collection within his cabinets, had |
| Emma.txt | except, perhaps, Mrs. Suckling's | own family, | and Mrs. Bragge's; but |
| Mansfield_park.txt | have it quite in their | own family | circle was what they had |
| Mansfield_park.txt | drawing-room, and all the | collected family, | were before her. As she |
| Mansfield_park.txt | a large and—he believed— | industrious family | was at stake. He had |

Through observation of the 20 lines from the output of the Austen corpus, the word "family" is more connected to social relationships and marriage background. For example, "respectable family", "ancient family", "fine family", "great family", and "large family", these terms in the context often reflected the respectability of social identity and how the family background impacts the marriage and social choices. Also, the word "family" frequently appears in contexts that express the intimate relationships of a family. Such as "own family", "whole family", "family party", and

“family circle”.

Compared to Dickens, Austen uses “family” less to discuss class conflict, focusing more on how family influences an individual’s marriage choices and social relationships.

***1c) Using the list of LancsBox grammatical smart searches, choose a second grammatical feature (other than ADJECTIVE) you think might be interesting and compare how it appears with your word (or a new one, or in general) in Austen and Dickens. Copy and, paste up to 20 lines of your output (there may be less) into your assignment doc, and describe any patterns you find.***

VERB[word="family"]

Dickens:

| | | | |
|---|---|---|---|
| Little_Dorrit.txt | the other hand, Miss Dorrit | had family; | and that her (Mrs Chivery |
| The_Life_And_ Adventures_Of_ Nicholas_Nickleby.txt | the wrist; ‘and don’t | carry family | matters to the neighbours, destroying |
| David_Copperfield.txt | Micawber, and our blighted but | rising family? | Need I say that this |
| The_Life_And_ Adventures_Of_ Nicholas_Nickleby.txt | to see a man’s | confounded family | conceit blinding him, even to |
| Little_Dorrit.txt | position here. A very well | conducted family | indeed, the Chiveries. Their behaviour |
| Little_Dorrit.txt | on the contributions of | changing family, | the greater stand he made |

| | | | |
|---|---|---|---|
| | his | | |
| Barnaby_Rudge.txt | much I dislike what are | called family | affairs, which are only fit |
| Our_Mutual_Friend.txt | upon myself when I have | respected family | friends to remember. Secondly, because |
| David_Copperfield.txt | and then, with my ill- | starred family, | to disappear from the landscape |
| David_Copperfield.txt | between Mr. Micawber and his | agonized family? | Oh no, for that would |
| The_Uncommercial _Traveller.txt | mother, she suckling the youngest, | discussed family | affairs as peaceably as if |
| Barnaby_Rudge.txt | are grave and weighty reasons, | pressing family | considerations, and apart even from |
| A_Christmas_Carol.txt | and endeavour to assist your | struggling family, | and we will discuss your |
| Bleak_House.txt | her away?" "She supposes we | have family | affairs to speak about, I |
| Barnaby_Rudge.txt | influence and rank--that he | had family | secrets in his possession--that |
| The_Old_Curiosity_ Shop.txt | It was rather a queer- | looking family: | Kit, his mother, and the |
| Bleak_House.txt | it be a mask--and | carries | secrets in |

| | | | |
|---|---|---|---|
| | | family | every limb of |
| The_Uncommercial _Traveller.txt | In dwelling-houses, there have | been family | festivals; children have grown in |
| Bleak_House.txt | so happily exemplifies the before- | mentioned family | likeness to the monkey tribe |
| A_Tale_of_Two_Cities_ A_Story_of_the_ French_Revolution.txt | his dead mother, on this | injured family, | if the sister can be |

Through observation of the 20 lines from the output of the Dickens corpus, the word “family” is not just family members, it also contains social functions and emotional attitudes. For example, “carry family matters to the neighbors”, “respect family members”, “have family secrets”, “pressing family considerations”, “struggling family”, and “discussed family affairs”, “family” is closely paired with verbs that relate to some behaviors. “Agonized family” and “struggling family” indicate life difficulties within the family. The family becomes a closed place where certain things can only happen in this space, like “have family secrets” and “carries family secrets”. Additionally, “family” also pairs with verbs related to social relationships and family rules, such as “respect family friends” and “discuss family affairs”; these terms indicate that family is not only a place where members live their personal lives, but also is used to maintain social relationships at that time.

“Family” more represents control over individuals rather than a warm place to protect individuals in the Dickens corpus.

Austen:

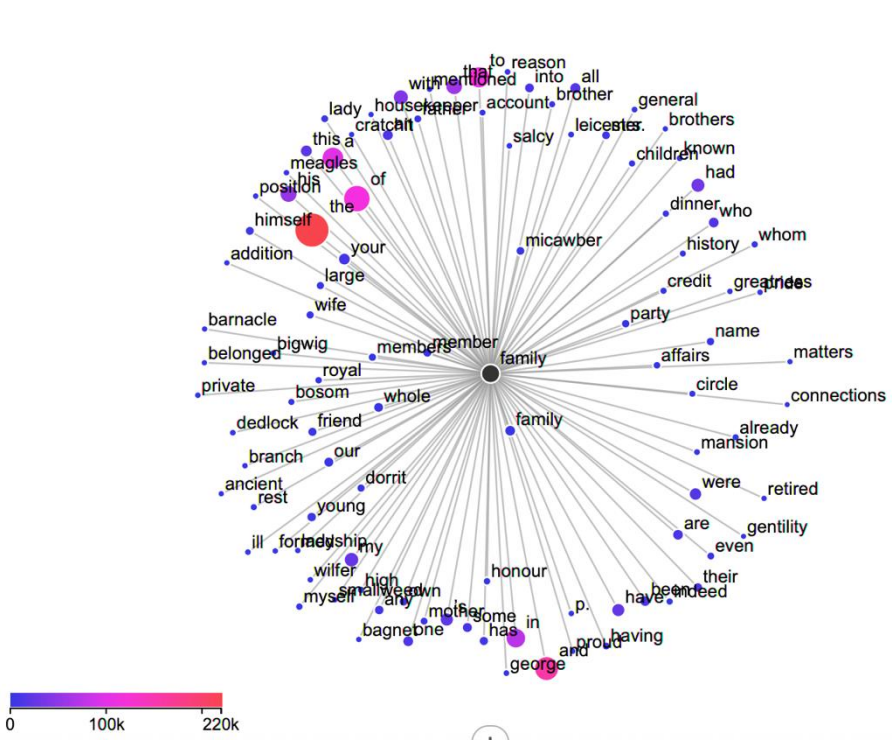
| | | | |
|---|---|---|---|
| Mansfield_park.txt | having so requited hospitality, so | injured family | peace, so forfeited his best, |

| | | | |
|---|---|---|---|
| Pride_and_prejudice.txt | we might possibly get a | settled family | there. But, perhaps, Mr. Bingley |
| Mansfield_park.txt | good, but the larger part | were family | portraits, no longer anything to |
| Mansfield_park.txt | her. A large and still | increasing family, | an husband disabled for active |

Through observation of the 4 lines from the output of the Austen corpus, the word "family" focuses on its stability. For example, "settled family", "increasing family", and "family peace" reflect the core position, continuity, and emotional function of a family in the marriage. Additionally, family is related to social status and inheritance, such as "family portraits", which imply family history and the importance of family as a symbol of class.

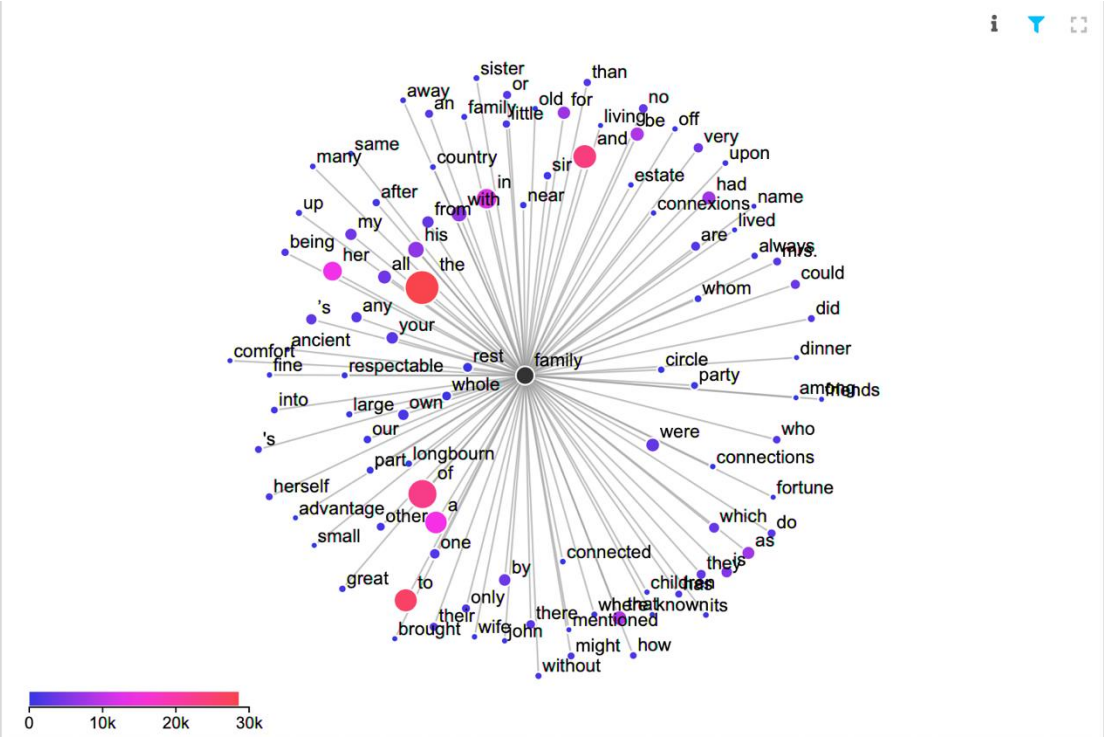
***1d) Using the GraphColl tool, find out what the clustering around your word looks like in the two corpora. Paste a screenshot of the output into your assignment. What are your observations?***

Dickens:




This picture indicates that “family” has a strong connection with person-referring words (such as himself, wife, brother, children), social status words (such as gentility, position, honor), and words related to family structure and social relations (such as circle, connections, affairs). Family is often placed into a certain context because of function words and designating words like “the”, “of”, and “his/himself”, etc., appear frequently with “family”. Family is important to an individual’s social identity, family status, and interpersonal connections. This word plays the role of kinship and narrative.

Austen:




"His", "her", "their", and "own" as some highly occurring words that pair with "family" indicate that "family" is often used to describe character affiliations and identities, emphasizing the role of family in social relationships. The presence of words such as "respectable", "ancient", "advantage", and "fortune" reveals a close connection between family and class, and wealth. In addition, words like "connections", "circle", and "part" show that "family" is often placed in the context of social interaction, reflecting the function of family networks in maintaining social and class structure.

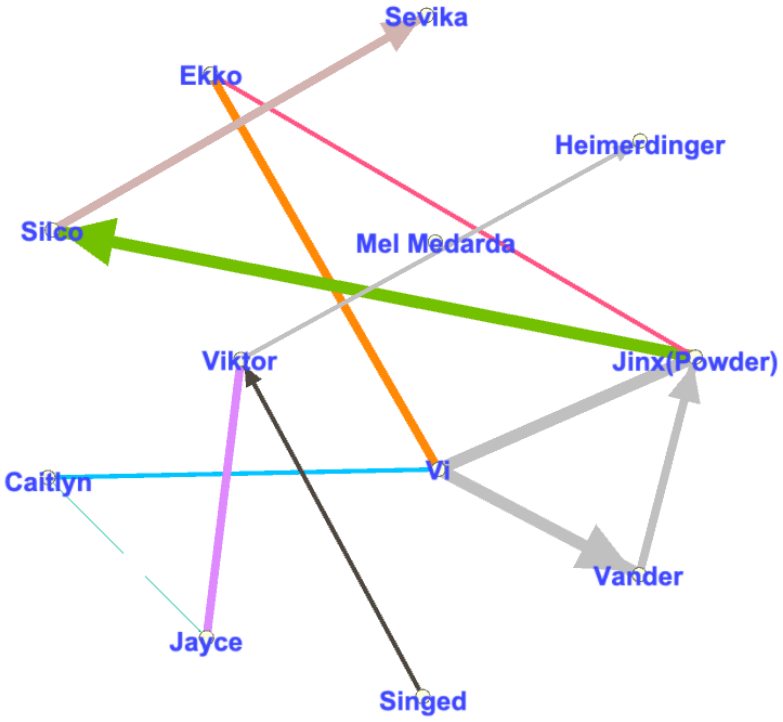
**Part 2: Continue with Gephi**

2a) Complete the edges.csv and nodes.csv files for 10-15 characters in a novel, TV series, film, video game, or any story of your choice (e.g., character interactions you know about in the English-language).

2b) Assign nodes as the characters and assign the weights and directions to the edges.

2b) Visualise in Gephi with Labels include a screenshot of your visualisation into your assignment.

2c) Add your analytical commentary into your Word Document (below Part 1: Lancsbox X). Consider in your analysis commentary: can you see any clusters, key nodes (i.e., characters) in the network?




By observing the above picture, the network clearly shows two major clusters: Vi, Jinx, Vander, and Ekko, from the Zaun faction. These characters share close relationships, often through family, friendship, or childhood friendships, with frequent and intense interactions. Another cluster mainly includes Jayce, Viktor, Heimerdinger, Mel Medarda, and Caitlyn from the Piltover/ Politics/ Technology faction, with edges reflecting scientific collaboration or political connections. Silco, Sevika, and Singed tend to be villains, serving as bridges that connect the two factions.

Regarding key nodes, Vi is at the center of the network, connecting Zaun and Piltover. Jinx provides a link between the upper and lower levels, while Silco controls the villain's power. Viktor plays a central role in the Science faction. Overall, the network

exhibits a centralized structure, with variations in edge weights reflecting the strength of interactions. The characters like Silco, Sevika, and Singed as bridge and cross-group nodes drive both conflict and cooperation in the plot.

**Reference list**

Austen Text Corpus. Corpus folder. 2025. Unpublished corpus, Trinity College Dublin.

Dickens Text Corpus. Corpus folder. 2025. Unpublished corpus, Trinity College Dublin.

Fortiche, and Riot Games. 2024. Arcane. Animated series. Netflix. Season 2. Video.